

Phase 1 - Week 4 - Lesson Day 1

Probability Basics

Today's Goal

Today we learn **probability** - a simple way to measure how likely something is to happen. This is the base for everything else we learn this week.

- Understand what probability means
- Learn simple probability rules
- Understand what a normal distribution is

1. What Is Probability?

Simple definition: Probability is a number between 0 and 1 that tells you how likely something is. **0** means it will never happen. **1** means it will always happen. **0.5** means it happens half the time.

The formula is simple:

```
probability = (number of ways it can happen) / (total number of possible outcomes)
```

Example: flipping a coin.

```
# A coin has 2 sides: heads and tails
# There is 1 way to get heads, out of 2 total outcomes

prob_heads = 1 / 2
print(prob_heads)    # 0.5 -> 50% chance
```

Example: rolling a dice.

```
# A dice has 6 sides (numbers 1 to 6)
# There is 1 way to roll a 4, out of 6 total outcomes

prob_rolling_4 = 1 / 6
print(prob_rolling_4)    # about 0.167 -> about 17% chance
```

2. Simple Probability Rules

Rule 1 - "AND" rule: if you want two separate things to BOTH happen, you multiply their probabilities.

```
# Probability of flipping heads TWICE in a row
prob_two_heads = (1/2) * (1/2)
print(prob_two_heads)    # 0.25 -> 25% chance
```

Rule 2 - "OR" rule: if you want EITHER one thing OR another thing to happen (and they can't both happen at once), you add their probabilities.

```
# Probability of rolling a 1 OR a 2 on a dice
prob_1_or_2 = (1/6) + (1/6)
print(prob_1_or_2)    # about 0.333 -> about 33% chance
```

3. What Is a Normal Distribution?

Simple definition: A normal distribution is a pattern where most values are close to the average, and fewer values are far away from the average. When you draw it as a chart, it looks like a bell shape - tall in the middle, short on both sides.

Example: most people's height is close to the average height. Very tall people and very short people exist, but there are fewer of them. That pattern is a normal distribution.

```
import numpy as np
import matplotlib.pyplot as plt

# Create 1000 fake heights. Average = 165, most values stay close to 165
heights = np.random.normal(165, 8, 1000)

plt.hist(heights, bins=30)
plt.title("Heights - Normal Distribution")
plt.show()
```

The number **165** is the average (center of the bell). The number **8** controls how wide or narrow the bell is - a bigger number means more spread out values.

Today's Practice

Open **day-1-probability.ipynb**. All answers are already filled in with simple code and clear comments. Run each cell and read the comments to understand what is happening.

Practice Questions (5 total - all solved)

1. Probability of a coin flip

Calculate the probability of getting tails on a coin flip.

2. Probability of a dice roll

Calculate the probability of rolling a 6 on a dice.

3. AND rule

Calculate the probability of rolling two 6's in a row (two separate dice rolls).

4. OR rule

Calculate the probability of rolling a 3 OR a 5 on one dice roll.

5. Plot a normal distribution

Create 500 fake test scores with an average of 70, and plot the histogram.

Once you and Halimo finish reviewing today's practice, move to Lesson Day 2 (Hypothesis Testing).